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CO2 -AT2

MLA0201-Fundamentals of Machine Learning

Q1. Linear Regression for Apartment Rent Prediction

Introduction

Linear Regression is a supervised learning algorithm used to predict continuous values. It predicts apartment rent based on floor area, number of rooms, and location.

Model

$$\text{��������} = \text{��}_0 + \text{��}_1(\text{��������}) + \text{��}_2(\text{����������}) + \text{��}_3(\text{����������������})$$

Where:

- Rent = Predicted apartment rent
- Area = Floor area
- Rooms = Number of rooms
- Location = Location rating

Steps

1. Collect apartment data.
2. Train the linear regression model.
3. Estimate regression coefficients.
4. Enter new apartment details.
5. Predict the rent.

Example

Area = 1200 sq.ft, Rooms = 3, Location = Good

The model predicts the monthly rent.

Advantages

- Simple and fast.
- Easy to understand.
- Suitable for price prediction.

Applications

- House price prediction
- Apartment rent prediction
- Sales forecasting

Conclusion

Linear Regression accurately predicts apartment rent using multiple features.

Q2. Linear Classification for Spam Detection

Introduction

Linear Classification is used to classify emails as **Spam** or **Non-Spam** using word frequencies.

Model

$$��(��) = ��_1��_1 + ��_2��_2 + ��$$

Decision Rule:

- $��(��) > 0 \rightarrow$ Spam
- $��(��) \leq 0 \rightarrow$ Non-Spam

Steps

1. Collect email data.
2. Extract important words.
3. Train the classifier.
4. Compute the decision function.
5. Classify the email.

Example

Offer = 2, Win = 1

If $��(��) > 0$, the email is **Spam**.

Advantages

- Fast.
- Easy to implement.
- Good for text classification.

Applications

- Spam filtering
- Fake news detection
- Sentiment analysis

Conclusion

Linear Classification helps automatically identify spam emails.

Q3. Naïve Bayes for Flu Prediction

Introduction

Naïve Bayes is a supervised learning algorithm based on **Bayes' Theorem**. It predicts whether a patient has flu using symptoms like fever, cough, and headache.

Formula

��(������ | ����������������) = ��(���������������� | ������) × ��(������)
��(����������������)

Steps

1. Collect patient data.
2. Calculate prior probability.
3. Calculate conditional probabilities.
4. Apply Bayes' theorem.
5. Predict whether the patient has flu.

Example

Symptoms:

- Fever = Yes

- Cough = Yes
- Headache = No

If ��(������ | ����������������)is higher than

��(���� ������ | ����������������), predict **Flu**.

Advantages

- Simple and fast.
- Works well with small datasets.
- Good for medical diagnosis.

Applications

- Disease prediction
- Spam filtering
- Document classification

Conclusion

Naïve Bayes predicts flu efficiently using symptom probabilities.

Q4. Discriminant Function for Loan Eligibility

Introduction

A Discriminant Function classifies loan applicants as **Eligible** or **Not Eligible** based on income and credit score.

Model

$$��(��) = ��_1(\text{Income}) + ��_2(\text{Credit Score}) + ��$$

Decision Rule

- $��(��) > 0 \rightarrow$ Eligible
- $��(��) \leq 0 \rightarrow$ Not Eligible

Steps

1. Collect applicant data.
2. Train the discriminant model.
3. Calculate the discriminant score.

4. Compare with the threshold.

5. Predict eligibility.

Example

Income = ₹60,000

Credit Score = 750

The model predicts whether the applicant is eligible for a loan.

Advantages

- Fast classification.
- Easy to implement.
- Suitable for binary classification.

Applications

- Loan approval
- Credit risk analysis
- Customer classification

Conclusion

Discriminant functions help banks quickly identify eligible loan applicants.

Q5. Probabilistic Generative Model for Handwritten Digit

Recognition Introduction

A Probabilistic Generative Model learns the probability distribution of handwritten digits and predicts the correct digit from an input image.

Model

��(���������� | ����������) =��(���������� |
����������) × ��(����������)
��(����������)

Steps

1. Collect handwritten digit images.

2. Train the generative model.

3. Learn probability distributions.
4. Input a new digit image.
5. Predict the digit with the highest probability.

Example

Input image = Handwritten digit

Output = Predicted digit (0–9)

Advantages

- Handles uncertainty well.
- Effective for image classification.
- Learns data distribution.

Applications

- OCR systems
- Bank cheque reading
- Postal code recognition

Conclusion

The probabilistic generative model accurately recognizes handwritten digits in OCR systems.

Q6. Probabilistic Discriminative Model for Customer Churn

Introduction

A Probabilistic Discriminative Model predicts whether a telecom customer will **churn** or **continue** using service data.

Model

��(��ℎ������ | ����������������)

Steps

1. Collect customer usage data.
2. Train the discriminative model.
3. Input customer features.
4. Calculate churn probability.

5. Predict churn.

Example

Features:

- High monthly bill
- Low usage
- Short service period

Prediction = **Churn** or **No Churn**

Advantages

- High prediction accuracy.
- Suitable for classification.
- Uses multiple customer features.

Applications

- Telecom
- Banking
- Customer retention

Conclusion

The model helps telecom companies identify customers who are likely to leave and take preventive actions.

Q7. Bayesian Logistic Regression for Product Purchase

Prediction Introduction

Bayesian Logistic Regression is a supervised learning algorithm used to predict whether a customer will purchase a product after viewing online advertisements. It estimates the probability of purchase using customer features.

Model

$$P(Purchase) = \frac{1}{1 + e^{-z}}$$

where

$$z = \beta_0 + \beta_1(\text{Ads Viewed}) + \beta_2(\text{Clicks})$$

Steps

1. Collect customer advertisement data.
2. Train the Bayesian Logistic Regression model.
3. Calculate the purchase probability.
4. Compare the probability with a threshold (0.5).
5. Predict **Purchase** or **Not Purchase**.

Example

Customer viewed **5 ads** and clicked **2 ads**.

The model predicts whether the customer will purchase the

product. **Advantages**

- Predicts probabilities accurately.
- Handles uncertainty.
- Suitable for binary classification.

Applications

- Online shopping
- Digital marketing
- Advertisement analysis

Conclusion

Bayesian Logistic Regression helps companies identify customers who are likely to purchase products after viewing advertisements.

Q8. Linear Regression for Monthly Sales Forecasting

Introduction

Linear Regression is used to predict monthly sales based on advertising expenditure and seasonal demand.

Model

$$\text{����������} = \text{��}_0 + \text{��}_1(\text{Advertising}) + \text{��}_2(\text{Season})$$

Steps

1. Collect previous sales data.
2. Train the Linear Regression model.
3. Estimate regression coefficients.
4. Enter advertising and seasonal values.
5. Predict monthly sales.

Example

Advertising Budget = ₹50,000

Season = Festival Season

The model predicts the expected monthly sales.

Advantages

- Easy to understand.
- Fast prediction.
- Suitable for forecasting.

Applications

- Sales forecasting
- Revenue prediction
- Business planning

Conclusion

Linear Regression helps businesses forecast sales and plan advertising strategies effectively.

Q9. Naïve Bayes for News Classification

Introduction

Naïve Bayes is a supervised learning algorithm used to classify news articles into categories such as **Sports**, **Politics**, and **Entertainment**.

Formula

��(����������������� | ���������������)

=��(��������������� | �����������������) ×
��(�����������������)
��(���������������)

Steps

1. Collect labeled news articles.
2. Extract important words from each article.
3. Calculate prior and conditional probabilities.
4. Apply Bayes' theorem.
5. Assign the article to the category with the highest probability.

Example

Article contains words like **match**, **player**, and **goal**.

The model classifies it as **Sports**.

Advantages

- Fast and simple.
- Good for text classification.
- Works well with large datasets.

Applications

- News classification
- Email filtering
- Sentiment analysis

Conclusion

Naïve Bayes efficiently categorizes news articles into the correct topic using word probabilities.

Q10. Bayesian Logistic Regression for Student Placement

Prediction Introduction

Bayesian Logistic Regression predicts whether a student will be **Placed** or **Not Placed** using CGPA, coding skills, and aptitude scores.

Model

$$P(\text{Placed}) = \frac{1}{1 + e^{-z}}$$

where

$$�� = ��_0 + ��_1(\text{CGPA}) + ��_2(\text{Coding Skills}) + ��_3(\text{Aptitude Score})$$

Steps

1. Collect previous placement records.
2. Train the Bayesian Logistic Regression model.
3. Enter student details.
4. Calculate the placement probability.
5. Predict **Placed** or **Not Placed**.

Example

Student Details:

- CGPA = 8.5
- Coding Skills = Good
- Aptitude Score = 80

The model predicts whether the student will be placed.

Advantages

- Accurate probability prediction.
- Uses multiple student features.
- Useful for placement analysis.

Applications

- Campus placements
- Recruitment systems
- Student performance analysis

Conclusion

Bayesian Logistic Regression helps colleges and recruiters predict student placement outcomes, enabling better career guidance and recruitment decisions.